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TWO

The Winning Strategy

THE DUTIES OF MARKETING are quite simple, yet few high-technology companies ever perform them. Why? Naïveté is one answer, but I suspect that in most cases the consequences of facing up to those responsibilities are so distasteful, companies refuse even to try.

For example, all companies should have the objective of being a leading supplier to their market. But few are willing to face their obligation to get out of a business if they don't achieve that goal. Instead, they wait to be driven out.

Other companies know they should target niches but hold to the hope of conquering the whole market with a "shotgun" approach. Still others would rather invent new products instead of performing the more arduous work on documentation, minor enhancements, and quality programs for existing ones. Only by such efforts will they find the great success they covet.

The fundamental truth of marketing comes down to this: If you are going to win the battle in the market place, you had better commit to the best strategy you can devise and implement it successfully. The market has no patience with sentiment. It rewards rational decisions executed with precision and conviction. Companies that succeed follow what might be called the "Strategic Principle" of marketing:

Marketing must invent complete products and drive them to commanding positions in defensible market segments.

If you are going to remember only one thing from this book, that principle and the reasons why it is true are it.

In the remainder of the chapter I shall elaborate on the strategic principle.

CRIMES IN THE NAME OF MARKET SHARE

In 1968 the Boston Consulting Group published its classic work on business strategies, *Perspectives on Experience*.* It was the book that introduced most of us to the notion of experience curves and to the need to maximize market share. (There was more to it than that, but no one seemed to notice.) Since then it has become fashionable to engage in the mindless pursuit of market share. Companies are keenly aware of the importance of holding commanding positions in their markets. But in an attempt to achieve those positions, too many commit themselves to reckless pricing policies, ill-conceived strategic plans, and ineffective marketing programs. After years of pursuing market leadership, many of those companies have nothing to show for their efforts but red ink.

That is not to suggest the BCG experience curve theory is wrong. As a matter of fact, much of the recent success of Japanese industry has been based on its application. What is wrong is not the theory but a lack of understanding of its application and the market. Companies fight for market share in the wrong market place with the wrong product. Or else they engage in wars of attrition against competitors able to commit more resources to the battle over extended periods. Both are recipes for defeat.

It is time to look again at what the Boston Consulting Group really had to say.

The BCG argued, based on a great deal of empirical data it had accumulated from such diverse industries as semiconductors, petrochemicals, primary metals, and consumer products, that the total cost of doing business decreased 20 percent to 30 percent every time business experience (production) doubled. Thus, if business costs

**Perspectives on Experience* (Boston: The Boston Consulting Group, Inc., 1968).

follow a 30 percent experience curve, which they quite typically do in the semiconductor industry, the total cost of getting a particular product to the customer should fall 30 percent every time unit sales double. For example, when the accumulated volume of a one-dollar device increases from one million to two million units, the cost should fall to seventy cents.

The BCG concluded:

> In a rapidly growing product area, the most successful competitive strategy is to achieve and hold a dominant market position either through pricing (or equivalent) tactics or by segmenting the market into a sufficiently isolated segment which can be dominated. If it is concluded that market dominance cannot be achieved, then an orderly withdrawal from the business is probably the best plan.

Unfortunately, most businesses choose to focus on strategies to achieve incremental positions in broad markets. They are far less enamored with "segmenting the market into a sufficiently isolated segment which can be dominated." Many have paid dearly for that blunder.

About the same time as BCG finished its study, General Electric completed an extensive survey of businesses to determine what constituted the basic elements of success for both emerging and mature businesses. To the best of my knowledge, that document has remained confidential, but some of the conclusions did leak out. One of them was that companies with greater than a 30 percent market share were almost always profitable, and companies with less than 15 percent almost always lost money.

The clear inference one can draw from both studies is that a company must achieve a commanding position in a market place or die. Furthermore, the number of companies that can survive profitably in any market place is limited. Certainly no more than six companies can have greater than a 15 percent market share. In fact, a review of those two and other studies leads to the obvious inference that most markets end up with only two or three consistently profitable suppliers.

That's a disturbing conclusion. Still, one need only look around to see a number of profitable automotive suppliers, many computer companies, and numerous semiconductor manufacturers. How do they survive?

The answer is that markets are not as homogeneous as they first appear. There is not one automotive market, there are lots of them: small cars, big cars, medium-size cars, sports cars, inexpensive cars,

luxury cars, imported cars, domestically manufactured cars, and on and on. Neither is there one wristwatch market. There are markets for disposable watches, watches worn as jewelry, fad watches, sport watches, mass-merchandised watches, and even scented watches. Similarly, for most classes of products there are numerous distinct market segments.

If those markets are distinct enough, then each can support two or three profitable suppliers. If they are large enough, a number of others can struggle along in them as well.

The point is that a company does not have to be as big as IBM to make a profit in the computer business, or as big as GM to make money in automobiles; *it only has to be large in its own protected market segment.* The market segment it serves may be isolated from competition by barriers to ward off competitive attacks, or it may survive because of the benign attitude of competition.

The problem most companies have in coming to grips with market share is in understanding market segmentation. The crimes committed in the name of market share are not a result of going for market share in and of itself. They result from pursuing market share in a haphazard fashion—usually in too broad a market.

The goal should never be only to capture a very small percentage of a huge market. Rather, concepts must be developed and resources identified to gain significantly greater than a 15-percent market share in a *well-protected* market segment. Anything else is to plan for failure.

DIVIDE AND CONQUER

Technology companies, when faced with massive competition, inevitably come up with the same tactic: to attack market segments, vertical markets, and niches. But, having proclaimed that strategy, they usually proceed as if the statement alone were all that is needed to succeed.

Market segmentation is one of the most discussed and least understood concepts in business. For that reason it is worthwhile relating to you one of my experiences in segmenting a market.

When I arrived at Intel in 1973, it was obvious that soon a number of potentially large businesses would be established based on the marketing of low-cost microcomputer systems. There already were several companies building hobby kits using microprocessors and selling them in reasonable volume.

I had my own vision of what the microcomputer market was going to be like. I imagined huge numbers of customers walking through the front door demanding to buy thousands of microprocessor chips.

The reality was significantly different. I was amazed by the number of customers trying to use micros in low-volume—many less than fifty chips a year—applications. The customers were having problems. Some couldn't make the hardware work, and others did not know how to write the software. This group of customers had a unique collection of needs. They represented a market segment.

What was even more interesting was that I found most of our potential competitors had the same skewed image of the market as I had entertained. They too thought most customers were buying micros by the thousands. (They probably had been reading Intel ads.)

So here was a big market segment walking in the door, begging for help, and being ignored by the industry.

As Bob Garrow (a top-notch systems engineer), Jim Lally, and I talked about that unrecognized market, we became more and more convinced it offered a terrific opportunity. We theorized about the ideal product for those people: the single-board computer. Our competitors had toyed with similar ideas, but none had really developed the type of system we were convinced the market needed. After all, we reasoned, microprocessors were only going to get cheaper. It was inevitable then that systems of the future would be crammed with them. If we could come up with a product that would make it easy for customers to use multiple processors in their systems, we would have a competitive edge.

There were, of course, some serious problems with the market. Digital Equipment Corporation was one of them. It was the world's leading minicomputer manufacturer and was working its way down into "our" product area. Although DEC did not yet have a true single-board computer, it had a competitive alternative.

Fortunately, DEC was not paying much attention to this new business; it was occupied with building bigger and better minicomputers.

Another problem with the market was its apparent low cost of entry. Engineering the types of devices we planned required little development effort. That meant, if we were successful, other firms would soon chase us.

To blunt the prospective attacks, we did several things. One move was to develop a new interface or "bus" structure to enable our customers to link multiple microprocessors into a single system.

"Multibus," as it was called, became an industry standard, and we freely licensed it to all would-be competitors. That was in direct opposition to DEC's strategy of "protecting" its bus structure from use by competitors. It helped to differentiate Intel from DEC, but it created myriad small competitors. However, we were confident we could deal with them. DEC was the real problem. As a result of licensing, Intel enjoyed a large number of add-on suppliers who expanded and enriched the product line and created some new niche markets. Further, our product was specifically targeted to the needs of Intel's existing customer base. Therefore, our sales and distribution channels were perfect for the job. DEC (and other competitors) did not have as good contact with the electrical engineering community, and that gave Intel a further advantage. Thus, when a microprocessor customer contacted Intel, and lots did, our salesperson could first find out if the prospect was a potential system customer and, if so, could attempt to sell a single-board computer.

Those were not our only advantages. Since Intel was designing components on one side of the house and systems on the other, we were able to make use of some of our newest chips long before they were available to the market—and to our competitors. Add to this Intel's brand image in the microprocessor market, and we had lots of big advantages. As the market developed, we differentiated ourselves even further by adding software, more proprietary products, and specialized distribution channels.

All of those things created barriers to market entry. As a result, the single-board computer business became large and profitable for Intel. The business, started with a $500,000 investment, long ago passed $100 million in revenues and enjoyed profit margins above the corporate average. Multibus's market share passed DEC's, and Intel is far and away the leading supplier of multibus products. Semiconductor companies, such as NEC, National, and Advanced Micro Devices (AMD), who raced Intel for the market, have dropped out. Intel now stands almost alone in this maturing market segment, protected by barriers difficult to cross.

That is what market segmentation is all about.

WHAT IS A MARKET SEGMENT?

Like many concepts in marketing, the notion of a "market segment" is deceptively simple. A market segment is a group of customers shar-

ing common desires, needs, and buying patterns. But in actual practice a marketing department that talks to enough customers will gather sufficient data to define enough market segments to guarantee confusion and paralyze any management team.

The real challenge in making market segmentation work is to identify the dominant characteristics in a customer population, then to create a product satisfying the needs arising from those characteristics. Much of the brilliant work done in consumer marketing and retailing results from keen insight, sometimes developed through painstaking research, as to what new products will have great appeal to a select customer group. Think of the Ford Mustang, the Swatch, and Miller Lite Beer.

Market segments are not neat and tidy. A single consumer can easily belong to many. Segments are the nonexclusive country clubs of marketing. The same consumer may enjoy shopping in both Neiman-Marcus and K-mart. Such bimodal behavior means that he or she won't fit neatly into any simplistic market analysis. Consumer migration from one segment to another can depend on things as simple as the length of time since the last paycheck.

WHAT CREATES MARKET SEGMENTS?

The forces segmenting markets are numerous. No single book could or should list all of them. What is important is understanding the types of customer preferences that go into creating segments.

Markets are most obviously segmented by product characteristics. High-tech companies tend to concentrate on that aspect. One group of customers may want to purchase only low-cost, "dumb" terminals; another, for reasons of productivity, may want high-performance, "intelligent" ones. Some customers for graphics work stations are happy with inexpensive low-power models, but Lucasfilms wants the ultimate graphics work station on which to fight the next *Star Wars*.

Capacity requirements are also important in segmenting markets. Customers needing large quantities of a product, such as General Motors, will do business with companies that have a high volume capability or have the financial resources to build it. The supplier unable to meet volume needs undermines that customer's business and quickly earns a reputation as a "boutique" manufacturer. The customer feelings that can develop as a result of supplier-caused shortages are very difficult to overcome.

In Japan, where much of the manufacturing is volume-oriented, suppliers tend to link up with a few very large customers. Their commitment is to the most-favored customers, often to the exclusion of all others. The less-favored customer in Japan is thus quite vulnerable. Unable to trust its source of supply, it seeks out other manufacturers. In that way buyer–seller relationships and customer wariness about those relationships create market segments in Japan (and, increasingly, here).

Quality, distribution channels, brand image, services, price, geography, level of documentation, customer comfort with corporate policy, customer needs for credit, the ability of users to relate to a supplier's management, and the flexibility of suppliers, their ability to alter capacity to meet customers' requirements, are some of the other factors segmenting markets.

The high-tech market for any device is not homogeneous. It is in fact tens of markets, many differing dramatically from the others. The individual market segments are varied and forever shifting. A product feature is meaningless one day and the next day all-important. Every difference in features and each shift in customer preference creates an opportunity for exploitation by an alert marketing team to gain a competitive advantage in a market segment.

HEAD ON IS HEADSTRONG

To access a market segment requires an investment as unique as the segment itself. That investment may be small, such as changing the color of a product. Or it may be quite large. A product line may have to be redesigned to meet new performance requirements. A new channel of distribution may have to be found.

The level of investment required of an outsider to enter a new market creates a *segment* or *entry* barrier. The segment barriers between two competitors will differ depending on their market position. If two companies serve similar segments, it may be relatively inexpensive for one to cross over into the other's market. For example, if two companies share many of the same distributors, as they do in the semiconductor business, the cost of attacking a competitor's market is significantly reduced.

Segment barriers become very large as an industry matures. If companies would only take the time to analyze the most important segment barriers facing them before they assault a competitor, a lot of

misery could be avoided. *Too much attention in business plans is focused on what a company plans to do, not on whether those plans are adequate relative to the complete set of segment barriers the company will face.*

Think of the barrier created by a big sales force. A digital system company sells approximately $1 million of products per salesperson per year. Achieving $1 billion in sales requires a sales force of a thousand salesmen, each costing the company about $100,000 a year. Therefore, at a billion-dollar sales level, the company is spending approximately $100 million a year on field sales. A small company with only ten salesmen and $10 million in revenues simply cannot compete with that. It must attack a niche.

The barriers to entry into soft drink markets can be very formidable as well. The advertising dollars required to build a nationwide brand image for a cola product are measured in the hundreds of millions of dollars. But even if one spends the money, the problems have just begun. A new competitor soon finds the biggest distributors in a given city already carry either Coca-Cola or Pepsi and aren't interested in bottling another brand.

Everywhere one looks, the companies that are profitable in their markets are protected by significant barriers. Find a good business, and you will undoubtedly find substantial barriers. For example, Tandem Computer supplies large, nonstop transaction processing computers. Its products are protected by massive hardware and software investments, in-depth application expertise, and a customer base whose confidence in Tandem is difficult to shake.

Segment barriers become strategically important when the cumulative cost of crossing them is large compared to the value of the market. When the cost of entering and gaining a position in a market becomes as high as the potential profits, then competitors are effectively barred. On the other hand, if the cost of crossing the barriers is only a small percentage of total market size, many will attempt to cross. A few will no doubt succeed. That situation exists most frequently when there is no dominant supplier to the market.

But even if the barriers are large, if the market is sufficiently attractive some companies will not be deterred. Technology companies are notoriously reckless when it comes to market entry. They tend to concentrate on the cost of developing and manufacturing a product rather than on the cost of crossing segment barriers. Thus many enter into a market chase naïvely.

What is the true cost of entering a market? I have a rule of thumb.

(To see how I developed it, see Appendix A.) The cost of entering a market against a well-managed competitor with an undifferentiated product—if such a thing can really exist—is 70 percent of the sales of the leader. From this it follows that the cost of crossing the segment barriers required to enter a market is:

$$.7 \times \text{(leader's share of market)} \times \text{(the size of the market)}$$

The formula explains why it is so difficult to enter the market when there is an entrenched leader. It also explains why companies such as Intel, Control Data Corp. (CDC), and DEC were able to enter large markets with very few resources. Simply put, they got there first, before an established competitor existed. Head-on assaults against entrenched competitors are usually suicidal. Conversely, once a company achieves a commanding position, competitors must make large investments if they hope to dislodge it with a head-on attack.

SEGMENTATION LETS DAVIDS SLAY GOLIATHS

Targeting a market segment makes participation less costly for a company and attack significantly less attractive for a competitor. To be a supplier to a broad market usually demands a broad product line. On the other hand, focusing on a segment means a leaner, tighter product family. That means reduced R&D costs.

Segmented markets also can be attacked with specialized distribution channels: A sales force of specialists can operate at significantly higher efficiency than the huge sales force of a broad-line competitor. Specialized distributors often have excellent market knowledge and contacts, which can be exploited. There may be a number of vertical trade shows and publications serving the customer base, which can be used economically to promote a product. Vertical promotions are far more cost-effective than horizontal ones.

Focus has other salutory effects. When a company commits to a narrow market, its best people can devote their attention to learning one segment, building their corporate image in the segment, and developing products to serve that precise customer base. Also, when the resources of a company are devoted to a specific market, the cumulative investment can become quite large compared to other competitors and to the market itself. For example, all advertising

dollars can be devoted to building brand acceptance in the segment. Specialized products and options can be developed to satisfy the unique needs of the customer base.

With all those advantages, a small company frequently can gain such a large share of a market segment that it makes no economic sense for a competitor to pursue the business.

By segmenting a market a small company can match the resources of a much larger competitor and at the same time build a defensible position. Segmentation makes it possible for companies with limited resources to build commanding positions.

IN HIGH-TECH YOU ARE NEVER SAFE

Now, after trumpeting the value of market segments, it's time for the bad news. There are several effective strategies to break down entry barriers. In a rapidly changing business you must always be on the lookout for a competitive attack. If you become complacent because of a big lead, someone will discover a good way to take your business away from you. The attacks will usually come from followers or from a market entrant using a new technology.

In commodity markets, playing the follower often works well. The Japanese have long used this strategy successfully. The GE study explains why. It identified two successful business strategies, one for leaders and the other for followers. The leadership strategy requires large investments both in R&D to develop new products and in marketing to develop the new market. Those costs seldom go away but live on, burdening product lines with high costs.

The follower does just the opposite. This firm copies a competitor or obtains a technology license. It spends little money on market development. Because the follower builds up its capacity after the leader, it can frequently take advantage of newer and more efficient manufacturing techniques. That's what the Japanese did in the steel industry.

The follower strategy in a commodity market gives a company low overhead, low manufacturing costs, and price leadership. Buyers are attracted by the opportunity to get a high-quality product in volume, on time, and at a bargain price. The strategy usually works well because market leaders allow it to. They are slow to react to the evolution of their market from specialty goods to commodities.

In other cases a single attack based on a new technology can destroy a leading supplier and reshuffle competition.

The semiconductor business provides many excellent examples of that effect. In the 1960s most computers used magnetic cores for their main memories. The suppliers to the market set up low-cost manufacturing facilities in Asia to string the tiny cores manually into arrays. Those firms constantly reduced the cost and improved the performance of their product.

But by the early 1970s, when Intel ran an ad with a headline, "The End: Cores Lose Price War to New Chip," it really was "the end" for most core suppliers. Semiconductor technology provided customers with faster and ultimately lower-cost memory technology. The core manufacturers' investments in design and manufacturing counted for nothing.

In slower-moving fields, the displacement of one competitor by another can take years. But in high-tech fortunes can change overnight. Companies at the forefront of technology truly balance on the razor's edge.

Over the years I have both won with technological breakthroughs and been the victim of them. I remember most vividly the defeats. I watched Intel's market share slip away in the development system business because of a change in technology. Many customers decided they wanted to do their work on large "super minicomputers" instead of on our microprocessor-based systems. Try as we might to deal with the problem, the effects of the shift were devastating. I watched dedicated followers, principally the Japanese, take market share from us in the mature components businesses. They executed their strategy well, and we had no effective response.

A company can lose a big lead if it fails to adjust its cost structure as its products evolve from specialty items to commodities or if it fails to exploit new technologies. Commanding positions do not ensure a permanent place in the market. The minute you think you are safe in a high-tech market segment, you have sown the seeds of your own defeat.

MYTHS ABOUT MARKET ENTRY

Companies are often deceived by markets that offer easy entry then turn out to be murderous to stay in. Often the cost of developing a

device to sell in a market is confused with that of establishing a leading position in that market. Companies forget that even with a good device they still haven't entered a market until they have established a significant position. Just as frequently, if a device is difficult to develop, management thinks the market must be difficult to enter.

Everyone knows how easy entry was into the PC business. For a few million dollars or less, a company could develop enough of a product to join the race. The truth is, massive markets require massive investments of resources. A small company such as Apple could successfully enter the market when the market was small and the barriers low. By getting in early, Apple was able to amass the necessary resources to have a large impact. The companies that followed could not. The market by then was more mature. In order to develop and sustain a position, the latecomers would have had to make investments equal to a significant proportion of the leader's annual sales. Few had such resources.

What it comes down to it this: There are very few markets where it is easy to achieve a commanding position after day one. Market entry is usually very difficult for late entrants.

GO FOR SEGMENT MARKET SHARE

In recent years a number of well-known companies, large and small, have vanished. That phenomenon has been most visible among high-tech start-up companies with initial success in the market place. Some made a little money, went public, then died.

There are thousands of reasons why companies go out of business, but a frequent one is a failure by a company to achieve a commanding position in a well-defended market segment. The shakeout phenomenon, which outsiders observe with morbid fascination, is the manifestation of this problem. Today it is the PC companies, yesterday it was the electronic calculator manufacturers, tomorrow it will be the computer-aided engineering vendors. *The companies that will vanish will be those that didn't make themselves unique in their customers' minds.*

Remarkably, many of the victims never even "planned" to succeed. They never appreciated the resources required to establish a leading position and maintain it. Frequently they attempted to retreat to a smaller market segment only after they had lost the bigger war.

And usually the retreat was accompanied by large writeoffs and debilitating layoffs. By then they were already doomed.

Many companies enter businesses planning to capture only a small percentage of the market. Such plans are not conservative; they are extremely dangerous. They are strategies for failure. The only good plans and the only good marketing programs are those aimed at dominance, backed by sufficient resources to achieve it, and executed with precision.

ENGINEERS INVENT DEVICES

The "miraculous" Intel electronic watch was a flop. The "wonderful" TI personal computer, even when pitched by the redoubtable Bill Cosby, failed. The problems were the same. The technologists created a good device, but marketing failed to invent a good product. Both firms tried to market something that satisfied only a portion of customers' needs. They built only partial products.

A true product is a very complex thing; it has a multitude of dimensions, all changing and evolving. Its features can be eroded by competition or transformed to pursue a shifting customer base.

Great devices are invented in the laboratory. Great products are invented in the marketing department.

Consider the invention of the microcomputer. In 1969 Busicom, a Japanese calculator company, visited Intel to discuss its idea for custom large-scale integrated (LSI) circuits to be used in a new calculator. Busicom wanted Intel to design twelve different circuits. Ted Hoff was heading Intel's applications effort at the time. Ted quickly realized there was no practical way to implement the devices proposed by the Japanese.

A computer man at heart, Ted was sure a small general-purpose computer could do the job. Further, he believed that everything could be fabricated from just three types of integrated circuits. That was the birth of the microprocessor.

I ran into Bob Noyce and Ted Hoff in the fall of 1969 at a computer conference in Las Vegas. We discussed future trends in integrated circuits. I described my concept for a "bipolar"-type circuit set that could do for high-speed applications what Ted's invention had done for less speed-intensive ones. A year later I joined a company that tried unsuccessfully to implement my idea. We had a good system design, but we lacked the requisite semiconductor technology.

In late 1971 I watched with envy as Intel announced the first microprocessor to the world. I now understood why Bob and Ted had been so interested in our Las Vegas conversation. Nevertheless, the stillborn product I was working on had some important implications for Intel, and I brought some of those ideas with me when I joined the firm in 1973. The key one was the concept of device emulation. That idea and many others were useful in augmenting Intel's product offering.

When I joined Intel I was stunned to learn how small the microprocessor business really was. In fact, Intel was selling less than $1 million a month of microprocessors, memories, and simulator boards. The market was slow to develop because of limitations in the devices and because most customers, frankly, didn't know what to do with them and couldn't get them to work in systems.

By 1974 a lot of new microprocessors were arriving in the market. Rockwell had the PPS-8; Motorola was ready to announce the 6800. Other companies had products in the wings. Intel's new device, the 8080, also made its debut that year.

In a comparatively short time the 8080 became the hands-down winner in the microprocessor market. The reason it won was not only that Ted Hoff and his technical team invented a superb *device* but because Intel invented the first broad market microprocessor *product*. One of the factors in turning the 8080 into a complete product was a device emulator based on concepts developed in my prior job.

The Intel 8080 was the most complete microcomputer product in its day. It was much more than the "chip." It comprised the "chip," to be sure, but also application notes, ads, microprocessor development systems and emulators, software, field applications engineering and single-board computers, customer education programs, and Bob Noyce's ability to capture the public imagination.

There were many inventors of the microprocessor. Certainly Ted Hoff and his crack technical team invented Intel's microprocessor families, but those were just devices. There were hundreds of other creative people who made contributions to the invention of the product. Intel had those people, and that is why it won the early microprocessor battles against other companies that entered the market with only devices.

In truth, thousands of devices are invented every year. Some, like my bipolar microprocessor, never see the light of day because of technical problems. Others emerge as only partial products, already as

doomed in the market place as if they had never made it off the engineers' desks.

That is not to denigrate the role of the engineer-inventor. All the marketing hype in the world can't create a product unless a device exists. But the key is to convert great devices into great products.

Everyone knows what a product is. It is what you buy at the store, the services of a doctor, or the advice of a consultant. Interestingly, the device often obscures the product. When shopping for food, the food (the device) is being purchased, but so are the retail environment, the display of the food, the cleanliness of the store, and the ease of parking (the product). People at a Rolling Stones rock concert are there to hear the group but also are buying a place in a social environment and the chance to brag to their associates afterward. A product is the totality of what a customer buys. It is the physical device or service from which the customer gets direct utility plus a number of other factors, services, or perceptions, which make the product useful, desirable, and convenient. When a *device* is properly augmented so that it can be easily sold and used by a customer it becomes *product*.

Incomplete or partial products are the misfits and deviates of the market place. They are the source of a great many high-tech marketing failures. (Henceforth we shall retain this semantic distinction between "devices" and "products.")

CREATING COMPLETE PRODUCTS IS EXPENSIVE

Complexity is the hallmark of high-tech products. They are more complicated than most consumer products. The complete product called a "computer," for example, consists of system hardware and a wide array of associated options, an operating system, application languages, application programs, documentation, customer training, salesman and presale application support, postsale application support, the maintenance organization and the logistic system that supports it with spare parts, the brand image of the company, the advertising and public relations about the system, and most importantly the feeling of confidence the customer has in the supplier. A company with only a few of those characteristics faces a problem similar to trying to sell an unpainted car. A customer will purchase it, but only at a very cut rate.

That complexity is perhaps the most fascinating aspect of high-technology products. The computer industry, still rapidly evolving, has provided a series of examples of the agony of partial products. Such large companies as GE, RCA, and Xerox tried building computer products but lacked an appreciation of what a complete product was. They failed in part because they underestimated what it would take to complete their computer system offering. For some applications their computers lacked software, for others they did not have the right kinds of peripherals. In many cases they were deficient in their ability to deliver high-quality product service. In almost all cases they lacked sufficiently broad sales coverage to create needs and solve problems for the customer.

GE struggled for years with unending losses. System after system was introduced, each claiming to be better than competitors' products. In spite of all efforts, customers did not buy GE computers in sufficient numbers. No matter what the company did, it always seemed to lack a product feature or service required by the customer.

In an attempt to broaden its product line, strengthen its peripheral product offering, and increase the capacity of its overseas distribution channels, GE purchased a French computer manufacturer and a portion of Olivetti. GE also narrowed its market focus (retreated to niches) with the hope of offering more complete solutions. It didn't work.

Finally, in the late 1960s, a frustrated GE appointed a strategic task force to study why the firm had not gained sufficient market share to be a viable competitor in the mainframe computer business. Reportedly the group concluded that each percentage point of market share gained would require an investment of more than $100 million. In the computer business the cost of completing a product is very large indeed. It is very high in most other businesses as well. *The cost of creating a complete product is often many times the cost of developing the device.*

DEFINING A HIGH-TECHNOLOGY PRODUCT

It's difficult to describe what a true high-technology product is. The very existence of most high-tech companies is predicated on developing new products and exploiting new businesses. Thus there are no well-worn paths to follow. One may not be able to find a prior group that did things almost right and left behind a business plan to be improved upon. It's also difficult to find failure models in a new field. They usually occur at the same time a leader is succeeding, often because of that success. It is thus almost impossible to study the mistakes made and the strategies and tactics that did not work. Role models are scarce.

One exciting new software business area is the "silicon compiler." These programs simplify and speed the design of integrated circuits. Few of these "devices" have been developed, and none are yet "products." To date, six silicon compiler companies have been founded, with more to follow. Each has a different strategy. Their proposed products are likewise different. It is anyone's guess what finally will be required to make a silicon compiler a complete product. Market research will be of little help, because potential customers won't know many of the subtle things they want until they get the product.

One of the leaders in the silicon compiler field is VLSI Technology, Inc. (VTI). It claims to offer its customers a complete cradle-to-grave service. The sales pitch is great, but the product is in no way complete. For some customers the human interface is too complex. Others want more extensive circuit libraries. For a few the software will not support their highly complex designs. Documentation and training are inadequate for another large portion of potential customers. Finally, the company's design centers are not convenient for many.

Is VTI doing a bad job? No. In fact, it is a leader in the field. But tens of millions of dollars more will have to be spent to complete the product. For VTI the most important objective is to achieve for its products a higher degree of completeness than that of any of its competitors' products. To do so VTI will have to segment the market. That won't be easy: It will not be obvious which segments are largest. Nor can the company be sure whether a target market can be developed if key parts of the finished product are still missing. Early users of any new and highly complex technology product almost always find some essential features missing. Will they stick around and wait for those additions to be completed?

A frequently made high-tech marketing mistake is to underestimate the number of dimensions a product must have. New devices are constantly being introduced to the market, yet technology companies continue to put their faith in good devices alone. They continue to ignore the importance of the complete product.

Most high-tech companies would be horrified to sell a device that did not work, but a large percentage seem able to tolerate devices that

cannot be used because of inadequate training or inaccurate documentation. The ultimate problem for the customer is the same: If the device can't be made to work, for whatever reason, it is useless. Similarly, high-tech companies are more concerned over the manufacturability of products than whether the distribution channels can sell them. Products bought by a distributor are not automatically sold to the customer. That lesson cost TI more than $100 million in the home computer business. Devices that don't work or can't be manufactured are incomplete products, but so are products that can't be successfully applied or sold.

A great deal of technical creativity goes into developing new devices. Far too little energy is expended on inventing complete products. The latter requires marketing innovation.

AN EVOLVING CUSTOMER BASE CHANGES THE PRODUCT

Customer needs and desires define products. As they change, so must the product. Evolving customers demand evolving products. During the life cycle of a product its customer base may change radically.

There is a well-known customer evolution: Innovators buy the first offering of a new high-technology product, then early adaptors and followers, and finally the late adaptors. The precise definition of those customer types is not important. What is important is that the device must be packaged as a different product for each group.

The "innovator" wants to be the first to use a new device. He will usually tell the salesperson his company is a technological leader and must stay ahead. He can't afford to wait; he even wants to be a test site. He delights in telling his suppliers' engineers just what is wrong with the new device or service. He wants to be part of the creation of the new product. For him a complete product is frequently a sadly incomplete one. He expects it to have poor documentation and many bugs. He is also willing to take the chance that the device may be delinquent and delay his own schedule.

On the other extreme, late adaptors, for psychological or sound business reasons, are not driven to use the latest innovations. Many are associated with businesses where the cost of mistakes is large and little is to be gained from risk-taking. Late adaptors are interested in complete and proven products and are frustrated by product malfunctions or errors in documentation. A good example of a late adaptor is an automobile company, which *must* use proven products in high

volume. To do otherwise would risk painful recalls and production line shutdowns.

The story of the personal computer business exemplifies how a market evolves. The innovators who purchased the first Altairs and Apples were mainly computer freaks who liked to play with the system and make it perform interesting tricks by programming it in arcane "machine language." Some early users were even sharp enough to modify hardware.

The real growth in the market came when application software became available for personal computers. The spreadsheet program Visicalc is sometimes described as the computer program that made Apple successful. That overstates the case, but Visicalc did enable Apple to sell many of its systems for business applications. Considerable amounts of application software written by independent software vendors (ISVs) turned the early Apples into complete products for the education market. Independent programmers wrote books about how to use the Apple, making up for the poor documentation that came with the product. Those contributions helped "complete" the Apple for certain market segments. Eventually the product evolved to the point where it was useful not only to innovators but to early adaptors, some followers, and a few late adaptors.

The personal computer sold today is a dramatically different product from the one Apple originally introduced. Interestingly, though the Apple became a much more mature product than when it was introduced and was used in a great many business applications, it was still vulnerable to a more complete product, the IBM PC. The PC was able to displace the Apple because of some important product features, excellent brand image, and strong distribution. The IBM label legitimized the PC for professional business applications. The PC became the complete product for business applications and turned the Apple into a device for that market.

Most high-tech products mature rapidly and in an agonizing fashion. A frequent mistake is to underestimate the degree of completeness required by the market. Often a company enters a market late with a device when measured against the existing market standards. The device might be a product for a market made up of innovators, but it will be inappropriate for the followers. So the product flops. Companies focus too much on the device, assuming that if it can just be made better success will inevitably follow. The truth is, *too much time is spent comparing one device to another and not enough comparing products.*

A product will achieve a commanding position within a market segment only if it is complete. Failure to complete a product leaves it vulnerable to attacks by competition. Marketing is the organization within a business charged with the responsibility for ensuring products are complete. It is also marketing's job to guarantee that the product remains complete over time as the customer base evolves and the competition sets new standards for completeness. That means regular reviews of every dimension of a product.

MARKETING INVENTS PRODUCTS

The Inventors' Hall of Fame contains the names of many creative geniuses. Surprisingly, few of those people ever invented a product. They certainly conceived of technological breakthroughs, and they are all great people. But all invented *devices*.

The Inventors' Hall of Fame does not recognize many creators of great *products*. Missing are Lee Iacocca, the inventor of the Ford Mustang and the New Chrysler Corporation, and Ray Kroc, who created McDonald's.

Those marketing triumphs were a result of creative thinking and hard work. They are the products of *marketing invention*. Once the device is sound, good marketing adds the rest. Promotional programs are developed to inform the customer about the product and to create interest. Services, such as application engineering and training, are added to ensure that the customer will achieve satisfactory use. New applications are discovered for the device, frequently not the ones envisioned by those who developed it. This demands as much inventiveness as the original experiments in the laboratory. Companies make big mistakes when they confuse device and product invention. Devices won't succeed over the long term in the market. That's why marketing invention is essential. It turns devices into products.

WHY HIGH-TECH MARKETING IS DIFFERENT

Marketing is characterized by diversity. Even marketing departments of companies in the same business may differ dramatically. Marketing functions in companies in different businesses will vary even more.

For example, in some industries salespeople call on customers. In

others they call on retailers. Distributors in the electronic components industry sell products directly to the consumers, while in many other industries distributors sell products only to retailers. In high-tech, products are usually sold to the customer by salespersons. In consumer products they are sold principally by advertising, packaging, and tasteful display.

Technology marketing differs from other types of marketing because of its emphasis on functions nonexistent or unimportant in other fields. In the computer business and related fields it is common to find companies spending more than 20 percent of revenues on direct sales, service, and postsale support. In most technology companies a tremendous amount of money is also spent on product documentation. By comparison, advertising and promotional expense amount to only a few percentage points of sales. In technology companies most of the marketing effort is devoted to direct sales, training and supporting distribution channels, customer education and application support, service, and postsale support.

In consumer businesses marketing budgets are allocated differently. Marketing puts considerable effort into designing packages and into advertising and promoting the product. Promotional budgets frequently run more than 10 percent of sales. Consumer marketing groups are expert at creating customer "pull." The customer is sold the product by promotions designed to create end demand. The salesperson calls on retail outlets and fights for shelf space, almost never dealing with the customer. Consumer product companies don't spend a great deal of their resources on postsale support, documentation, and customer education. Instructions aren't needed with tubes of toothpaste or cigarette packs. The documentation costs for a bottle of aspirin are quite small.

The decision to purchase a consumer product is usually a low-risk one, as most are relatively inexpensive. If a customer is not happy with a purchase, it can be returned or thrown away. For inexpensive products, consumer awareness is key to sales. A good advertising campaign can motivate customers to seek out a product and buy it.

In contrast, technology products are almost never sold by advertising. At best, advertising only raises customer awareness and creates a desire to learn more about the product. That's because the purchase of a high-tech product is often a high-risk decision. When a customer is going to spend thousands and sometimes hundreds of thousands of dollars, impulse buying is rare.

Often the truth about a high-technology product is not known un-

til months after it is purchased. Some pieces of capital equipment can't really be evaluated for years. That is true of much of the equipment used to build advanced semiconductors, as well as of computer systems. A company making a large data processing installation may spend much of a decade getting it to work properly.

When customers make decisions like those, the intangible factors become important. Customers evaluate suppliers on how well they will perform and on how much they can be trusted. Customers ask of a potential supplier: Will you really fix the software bugs? Will you really be available to service my equipment in two hours? Will you really be in business five years from now? Will you really complete work on the software modification on time?

The market for many new high-tech products is becoming increasingly consumer-like. That has prompted many high-tech companies to employ numbers of consumer marketing types. They have frequently failed when they have strictly applied consumer marketing techniques. The reason is that the application of those techniques is insufficient to complete the product. Technology products can benefit greatly from consumer techniques, but by themselves they are not enough.

Advertising helps to create trust but is only a small part of the selling process. Reputation, service, support, and references are more important. Those and many other factors make marketing high-tech different.

PRODUCT COSTS ARE DOMINATED BY MARKETING

In more and more technology businesses, the price of the product is dominated by the cost of marketing. In personal computers the cost of distributing a product is frequently greater than 40 percent of the sale price. Computer retailers receive 65 percent gross margins on product options. The advertising and marketing expense can run another 10 percent. Meanwhile the cost of actually manufacturing the product is often less than 25 percent of the recommended sale price. Thus distribution and marketing costs can be more than twice the manufacturing cost. In the mainframe computer business the costs of selling, supporting, and educating the customer are approximately equal to the cost of making the equipment. For the early microprocessor, customers were so thirsty for information that the cost of providing them with "free" manuals actually exceeded the cost of making the product. Software products cost almost nothing to make.

During its life cycle, the cost of completing a product can change dramatically. For example, some products become commodities in a relatively short time. For them the cost of marketing usually falls. Other products, sold to an increasingly broad application base, may require more and more support. There, the cost of marketing will increase as the product matures.

For the near future at least, the costs of manufacturing increasingly complex devices, driven principally by advances in microelectronics and automation, are going down, while the costs of distribution and service are heading up. The customer, of course, is concerned only with the cost of the product, not with whether the price is driven by manufacturing or marketing costs. The firms that succeed will be those paying attention to getting complete products to the customer in a more cost-effective fashion. For many products, squeezing the last penny out of manufacturing may not be as important as ensuring efficient operation of the sales and marketing functions. Many products won't be able to achieve commanding positions in the market unless they are marketed in a cost-effective fashion. They will end up costing too much.

DECISIVE COMPETITIVE ADVANTAGE

When marketing has done its job right, the company will enjoy decisive competitive advantages. It owns its turf. If a competitor sets foot on it, an aggressive marketing organization will be able to drive that intruder away.

The reason for segmenting the market and inventing complete products is to gain superiority in the market place. The device a company sells may not be particularly superior to the competition's. The device itself may be a little bit better technically or even slightly inferior when a customer closely examines the specifications, but the complete product will be so much better overall that no competitor can effectively challenge it.

Technical deficiencies, if they exist, can be overcome by superior distribution, better service and support, and a whole series of intangibles. Competitors will find themselves frustrated trying to overcome high levels of trust and long-term relationships.

The totality of what a leading supplier with a complete product does for his customer is so overwhelming that a few technical deficiencies will be overlooked. When companies offer complete products to a

loyal customer base, the customer will be so satisfied with the product that alternatives will be evaluated with great reluctance. Even when a competitor comes up with a product offering that is in many ways more attractive, the customer will give the premier supplier the benefit of the doubt.

At the end of the battle for a market segment, the company having done the best job is entitled to a secure lead, a lead that will let it earn a fair return on its investment and then the chance to continue doing a superior job for the customer. It's a nice position to be in.

THE STRATEGIC PRINCIPLE

All that I have just said can be distilled into a single marketing maxim. To repeat:

> **Marketing must invent complete products and drive them to commanding positions in defensible market segments.**